Applied Human Computer Interaction (AHCI)

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Chapter 1 – The Human

Activity: *Design Rage Poll*

Who has ever been frustrated by:

- A confusing app?
- A bad ATM interface?
- A website that works on desktop but not mobile?

Activity: *Design Rage Poll*

1. Poorly designed ATM screen

Too many options, Unclear buttons, Confusing wording or layout

2. Overcrowded mobile app UI

Tiny text, Too many icons, No clear primary action

3. Confusing COVID registration portal

Long forms, Unclear steps, Contradictory instructions

Activity: *Design Rage Poll*

Find **ONE** badly designed app or system from your phone and share below things:

You must answer:

1. Who is the user?
2. What is the main task?
3. Where will the user struggle?
4. How can the interface be improved?

Discussion Time

AHCI is about designing systems for real humans, not ideal users.

Humans are imperfect, they forget, they make mistakes, they use systems under stress

Real-World Examples:

1. Google Search autocomplete
2. Netflix recommendations
3. ChatGPT interface ATM confirmations (“Are you sure?”)

Transition to AI: Modern interfaces don’t just respond, they adapt.”

Why This Matters?

- User research
- Personas
- Prototyping
- Usability testing
- AI-driven interfaces
- Ethics in AI-HCI

Road Map to Careers

- UX Engineer
- Product Manager
- AI Product Designer
- Startup founder
- Research roles

“Every successful product is a successful HCI product.”

Assessments

- Mid Terms
- Finals
- **Project**
- **Class Activities & Readings**

Course Book: Human Computer Interaction 3rd Edition - ALAN DIX, JANET FINLAY, GREGORY D. ABOWD, RUSSELL BEALE

What Exactly is HCI?

Human–Computer Interaction (HCI) is the study of how humans interact with computers and how we design systems that are **easy, efficient, and satisfying** to use.

Chapter 1: The Human

Why the “Human” Comes First

- Computers are built for humans
- Users are the reason systems exist
- If users fail, systems fail

Before designing systems, we must understand **human capabilities and limitations**

Fast algorithm ≠ usable software
Feature-rich app ≠ user-friendly app

Psychology in HCI (Not a Far Thing!)

- Users think
- Users remember
- Users get confused
- Users make mistakes

Example: Why users forget passwords but remember Instagram gestures.

Cognitive Psychology

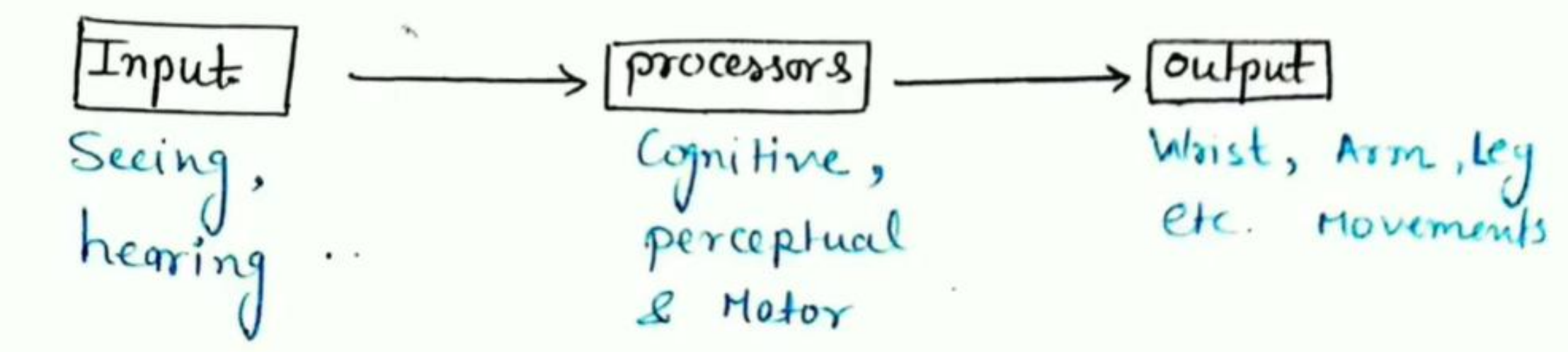
Cognitive psychology studies:

- How humans **perceive**
- How they **remember**
- How they **think**
- How they **solve problems**

All of these affect software usage

Model Human Processor (1983) - Card, Moran & Newell

- Simplified model of human interaction with computers
- Helps designers predict user behavior



Components of the Model Human Processor

1. Perceptual System

- Sees, hears, feels

2. Cognitive System

- Thinks, reasons, decides

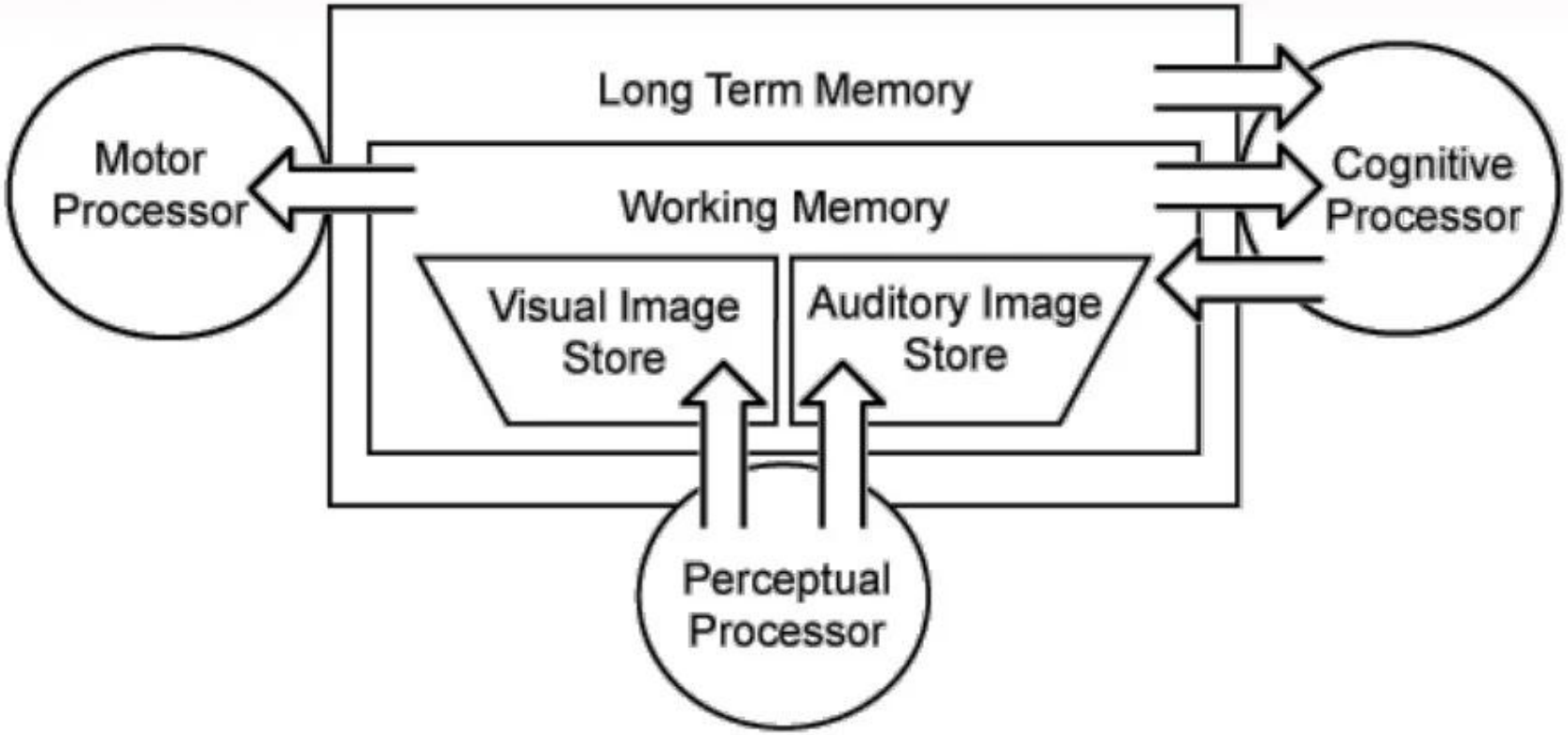
3. Motor System

- Hands, fingers, voice

Instagram Example

Action	System Used
Seeing icons	Perceptual
Deciding to like	Cognitive
Tapping screen	Motor

Model Human Processor (MHP)



Input–Output Channels in HCI

Humans interact with the world through input and output

In HCI:
Computer → user (output)
User → computer (input)

These roles often overlap **The same channel can act as both input and output.**

Example: Looking at a screen → input to human

Touching screen → input to computer

Human Input – The Senses

Five Senses: Vision, Hearing, Touch, Smell, Taste

HCI Focus (Important): Vision, Hearing, Touch

Examples:

- Vision → Instagram UI
- Hearing → Notification sounds, error beeps
- Touch → Keyboard feedback, phone vibration

Human Output – Effectors

Effectors (Output Channels): Fingers & hands, Voice, Eyes (eye tracking), Head & body position

Most Used in HCI: Keyboard, Mouse / Touchpad, Touchscreen, Voice (Alexa, Google Assistant)

Example:

- Typing code → finger motor control
- Saying “Hey Google” → vocal effector
- Touch gestures → mobile apps

Vision in HCI – Why It Matters

Vision = main source of information for users

Human vision has physical + perceptual limits

Interface design must respect these limits

Example:

- Dark mode vs light mode in apps
- Small fonts in coding IDEs cause eye strain

Two Stages of Visual Perception

Stage 1: Physical Reception

- Light enters the eye
- Image focused on retina

Stage 2: Interpretation

- Brain fills missing information
- Meaning is constructed, not copied

HCI Example:

- Loading skeleton screens (brain fills gaps)
- Blurred thumbnails still recognizable

Interpreting the Signal – Size & Depth

Visual Angle

- Indicates how much of the view an object occupies
- Depends on object size + distance from eye

Visual Acuity

- Ability to perceive fine detail (limited)
- Normal vision:
  (Detect single line at 0.5 seconds of arc)
  (Detect gaps at 30–60 seconds of arc)

Key Point: If visual angle is too small → object is not perceived

Examples:

Small text unreadable on mobile screens

Thin lines in graphs hard to see

Icons too small on dense dashboards

Interpreting the signal (cont.)

- Brightness
  - subjective reaction to levels of light
  - affected by luminance of object
  - measured by just noticeable difference
  - visual acuity increases with luminance as does flicker
- Colour
  - made up of hue, intensity, saturation
  - cones sensitive to colour wavelengths
  - blue acuity is lowest

The capabilities and limitations of visual processing

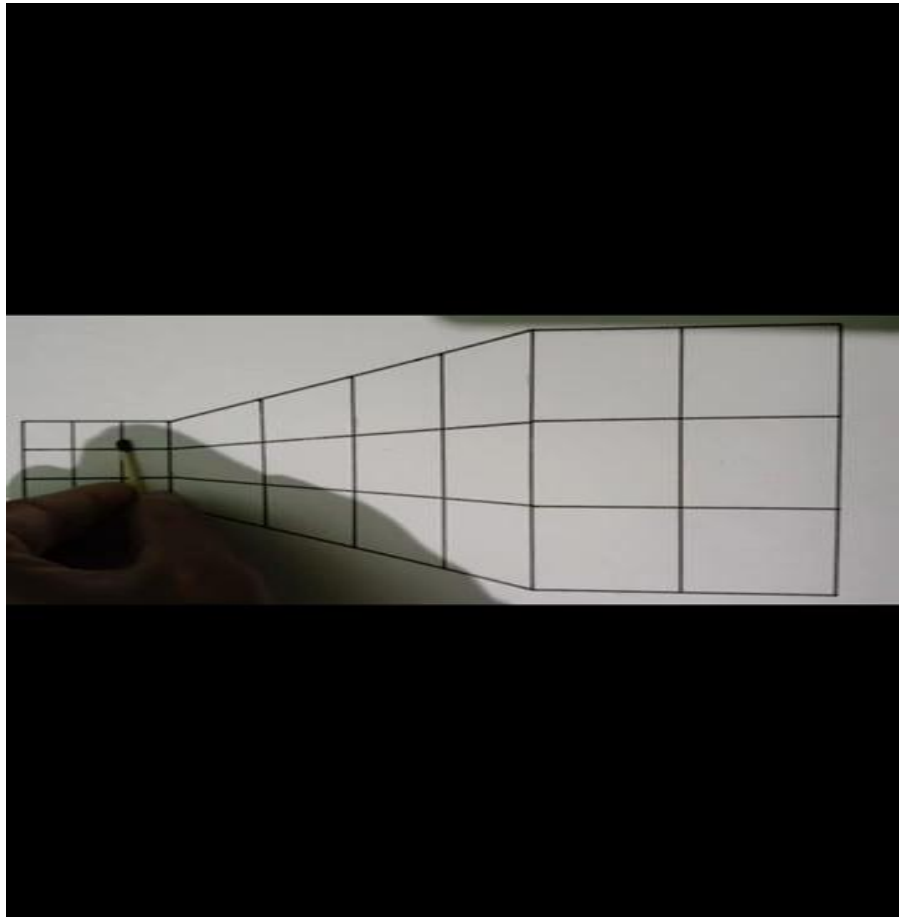
- Visual processing involves the transformation and interpretation of a complete image; For example, if we know that an object is a particular size, we will perceive it as that size no matter how far it is from us.
- This ability to interpret and exploit our expectations can be used to resolve ambiguity.

Figure 1.3 An ambiguous shape?

Figure 1.4 ABC

Figure 1.5 12 13 14

Ponzo illusion



Reading as a Visual Process

Reading Involves Several Stages:

- Visual pattern perceived
- Decoded using internal language model
- Interpreted using:
- Syntax
- Semantics
- Pragmatics

Example:

- Reading code comments
- Understanding variable names

Saccades & Fixations

- Reading involves:
  - **Saccades** → rapid eye movements
  - **Fixations** → pauses on words
- Actual perception happens **during fixations**
- Users don’t read continuously, they scan & pause

Examples:

- Skimming documentation
- Scanning Stack Overflow answers

Hearing

- Provides information about environment: distances, directions, objects etc.
- Physical apparatus:
  - outer ear – protects inner and amplifies sound
  - middle ear – transmits sound waves as vibrations to inner ear
  - inner ear – chemical transmitters are released and cause impulses in auditory nerve
- Sound
  - pitch – sound frequency
  - loudness – amplitude
  - timbre – type or quality

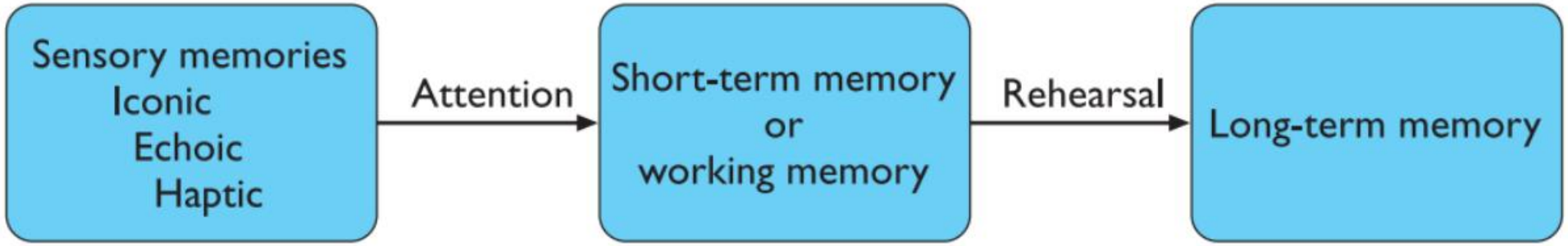
Touch

- Provides important feedback about environment.
- May be key sense for someone who is visually impaired.
- Stimulus received via receptors in the skin:
  - thermoreceptors – heat and cold
  - nociceptors – pain
  - mechanoreceptors – pressure
    (some instant, some continuous)
- Some areas more sensitive than others e.g. fingers than head.
- Kinesthesis - awareness of body position
  - affects comfort and performance.

Movement

- Time taken to respond to stimulus: reaction time + movement time
- Movement time dependent on age, fitness etc.
- Reaction time - dependent on stimulus type:
  - visual ~ 200ms
  - auditory ~ 150 ms
  - pain ~ 700ms
- Increasing reaction time decreases accuracy in the unskilled operator but not in the skilled operator. E.g. skilled typist vs novice.

Memory


Sensory memories
Iconic
Echoic
Haptic
Attention
Short-term memory
or
working memory
Rehearsal
Long-term memory

Short-Term Memory (STM)

- Acts as a temporary scratch-pad
- Stores information needed briefly
- Used for: Calculations, Reading comprehension, Decision making

Examples:

- Mental calculation (35 × 6)
- Remembering start of a sentence while reading
- Holding an OTP before entering it

Capacity of Short-Term Memory

- STM Capacity – 7 ± 2 Rule
- Average capacity: **7 ± 2 items**
- Established by **George Miller**
- Applies to: Digits, Letters, Symbols

Example:

- Remembering a 6-digit OTP
- Failing to recall a 12-digit number

Chunking

Chunking Information

- Grouping items into meaningful units
- Increases effective STM capacity
- STM remembers **7 ± 2 chunks**, not items

Example:

- 265397620853 (hard)
- 44 113 245 8920 (easy)
- Phone numbers grouped as 3-3-4

Design Focus

Cashing in

Closure gives you a nice 'done it' when we complete some part of a task. At this point our minds have a tendency to flush short-term memory in order to get on with the next job. Early automatic teller machines (ATMs) gave the customer money before returning their bank card. On receiving the money the customer would reach closure and hence often forget to take the card. Modern ATMs return the card first!

Recency Effect in Short-Term Memory

In experiments where subjects were able to recall words freely, evidence shows that recall of **the last words** presented is **better than recall of those in the middle**

This is known as the recency effect.

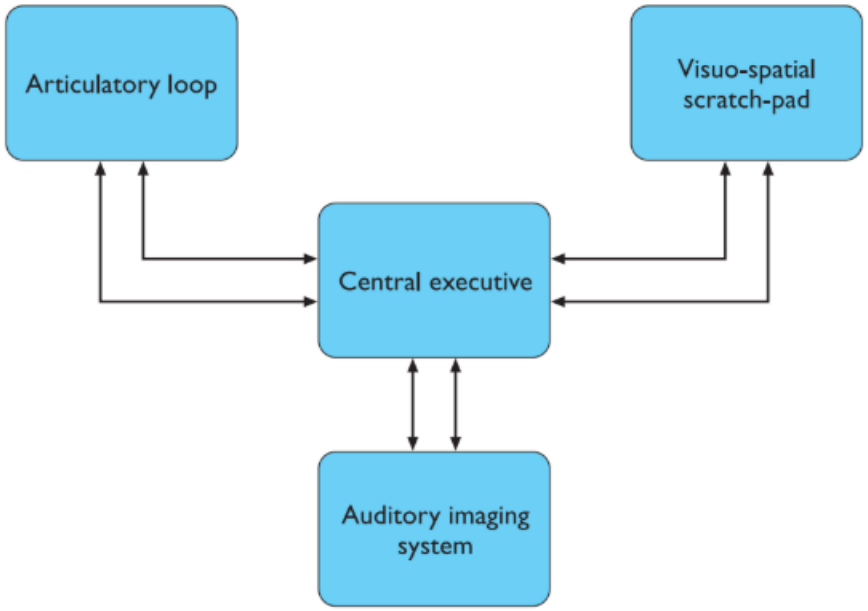
The **recency effect** means that **we remember the last things we see or hear better than the things in the middle**.

Examples:

- You remember the last message someone sent, but forget earlier ones
- While shopping, you remember the **last item** you added to the list

Baddeley's Working Memory Model

Baddeley's working memory model explains that short-term memory consists of multiple components controlled by a central executive, allowing parallel processing of visual, verbal, and auditory information



Central Executive – The Manager

What it does: Controls attention, Decides **what to focus on,** Switches between tasks, Sends work to the right helper

While coding, deciding whether to: Read error message or continue typing code

Articulatory Loop – Words & Numbers Helper

What it handles:

- Spoken words
- Numbers
- Repeating information in your head

Remembering:

- Variable names
- Function names
- OTP while typing it

Visuo-Spatial Scratch Pad – Visual Helper

What it handles: Images, Shapes, Layout, Space & position

Example: Remembering where you parked your bike, Visualizing a room layout

Auditory Imaging System – Sound Helper

What it handles: Sounds, Music, Audio patterns

Example: Remembering a song tune, Recognizing a ringtone

Long-Term Memory (LTM)

- Long-Term Memory (LTM) = **main storage of human knowledge**
- It stores facts, experiences, procedures
- **Huge capacity** (practically unlimited)
- **Slow access** (~0.1 seconds) compared to working memory
- Forgetting is **slow or minimal**

Example: Keyboard shortcuts, How Git works (commit → push → pull)

HCI Insight: Good UI reduces memory load by leveraging what users already know

How Information Enters Long-Term Memory

- Information moves from Working Memory → LTM via rehearsal
- Unlike working memory:
  No rapid decay
  Recall after minutes ≈ recall after days
  Repetition + meaningful use = stronger memory

Example

- Repeatedly debugging code → you remember common errors
- Using the same app UI daily (Instagram, VS Code) → actions become automatic
- HCI Insight:
  - Interfaces should support **learning by repetition**
  - Consistent UI patterns = faster user mastery

Types of Long-Term Memory

1. Episodic Memory

- Memory of **events & experiences**
- Stored in **sequence**
- Personal and contextual

2. Semantic Memory

- Memory of **facts, concepts, rules**
- Structured, organized knowledge
- Built from episodic experiences

Long-Term Memory Cont.

- **Semantic memory structure**
  - provides access to information
  - represents relationships between bits of information
  - supports inference

  Model: semantic network
  - inheritance – child nodes inherit properties of parent nodes
  - relationships between bits of information explicit
  - supports inference through inheritance

Semantic Memory & Semantic Networks

A **Semantic Network** is a way of representing **knowledge in long-term semantic memory** as a **graph-like structure**, where:

- **Nodes** represent **concepts or objects**
- **Links** represent **relationships** between concepts
- This structure helps humans **store, retrieve, and infer information efficiently**.

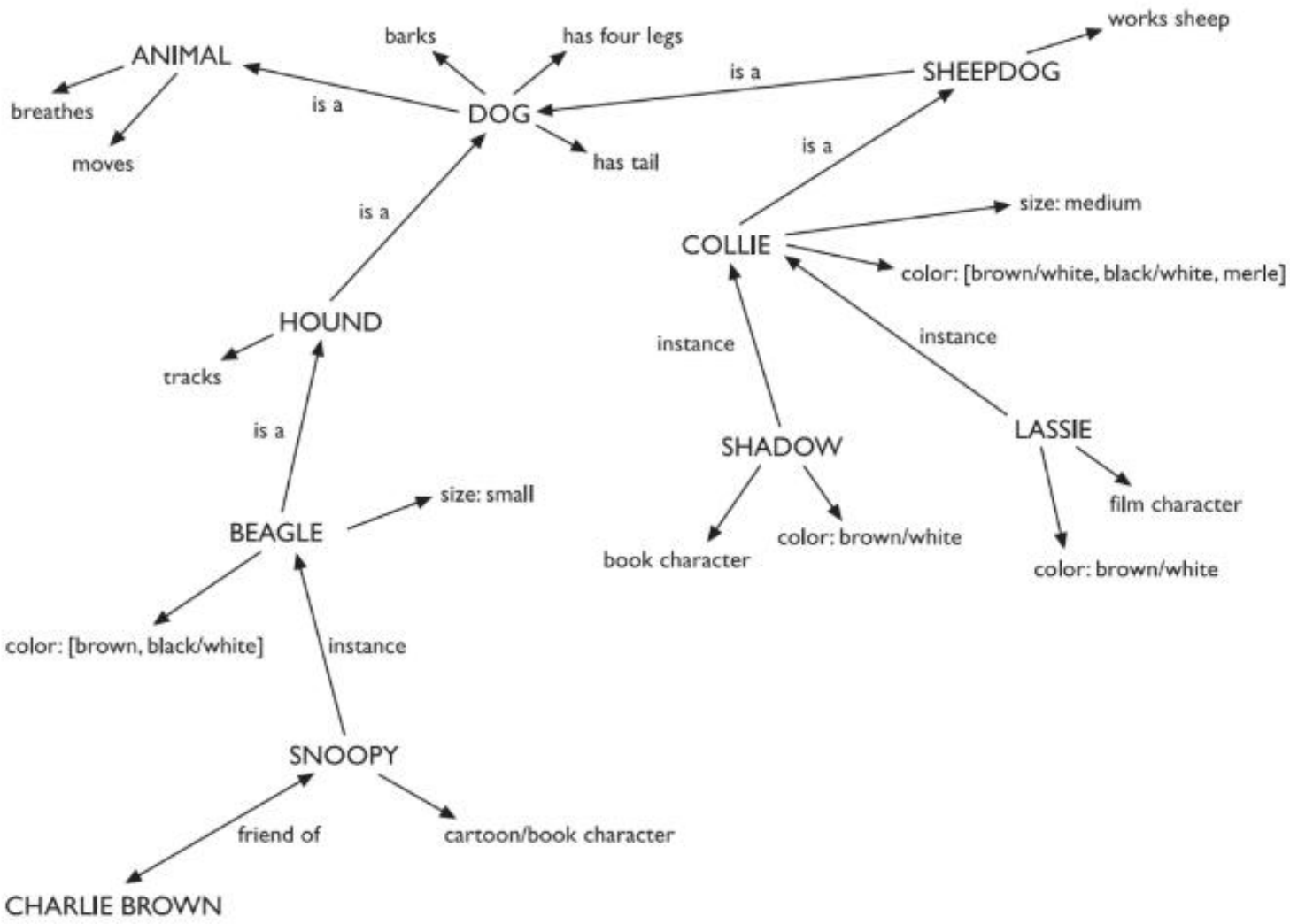


Figure 1.11 Long-term memory may store information in a semantic network

Semantic Memory & Semantic Networks

1. Nodes (Concepts) – Animal, Dog, Collie, Lassie
2. Links (Relationships)

IS-A → class/subclass relationship
Dog IS-A Animal
HAS-A → attribute relationship

Dog HAS-A tail

INSTANCE-OF → specific example

Lassie INSTANCE-OF Collie

Semantic Memory & Semantic Networks

Inheritance of Properties:

- Child nodes inherit properties from parent nodes
- Reduces memory storage and speeds up retrieval

Example from Figure

- Animal → breathes, moves
- Dog inherits → breathes, moves
- Collie inherits → tail, four legs

Why Semantic Networks Are Efficient

- **Structured storage** instead of random memory
- **Fast access** to related information
- Supports **inference**

Example

If Dog IS-A Animal

Animal breathes

Then: Dog breathes (even if not explicitly stored)

Scripts – Knowledge About Typical Situations

- A **script** represents **stereotypical knowledge** about **common situations** and **events**. Scripts attempt to model the representation of stereotypical knowledge about situations.

Example Sentence: “John took his dog to the surgery. After seeing the vet, he left.”

Even though many steps are not mentioned, we **assume**:

- Dog was ill
- Vet examined the dog
- Treatment happened
- John paid before leaving

This happens because we have a **“visit the vet” script** stored in long-term memory.

Elements of a Script

Script for a visit to the vet

Entry conditions:	*dog ill* *vet open* *owner has money*	Roles:	*vet examines* *diagnoses* *treats* *owner brings dog in* *pays* *takes dog out*
Result:	*dog better* *owner poorer* *vet richer*	Scenes:	*arriving at reception* *waiting in room* *examination* *paying*
Props:	*examination table* *medicine* *instruments*	Tracks:	*dog needs medicine* *dog needs operation*

Figure 1.13 A script for visiting the vet

Thinking & Problem solving

- Deductive Reasoning: Derive logically necessary conclusion from given premises.
- Example: If it is Monday then he will go to work
- It is Monday, Therefore, he will go to work.

Logical conclusion, but not necessarily true:

- Example: If it is raining then the ground is dry
- It is raining, Therefore, the ground is dry **Correct?**

Inductive Reasoning

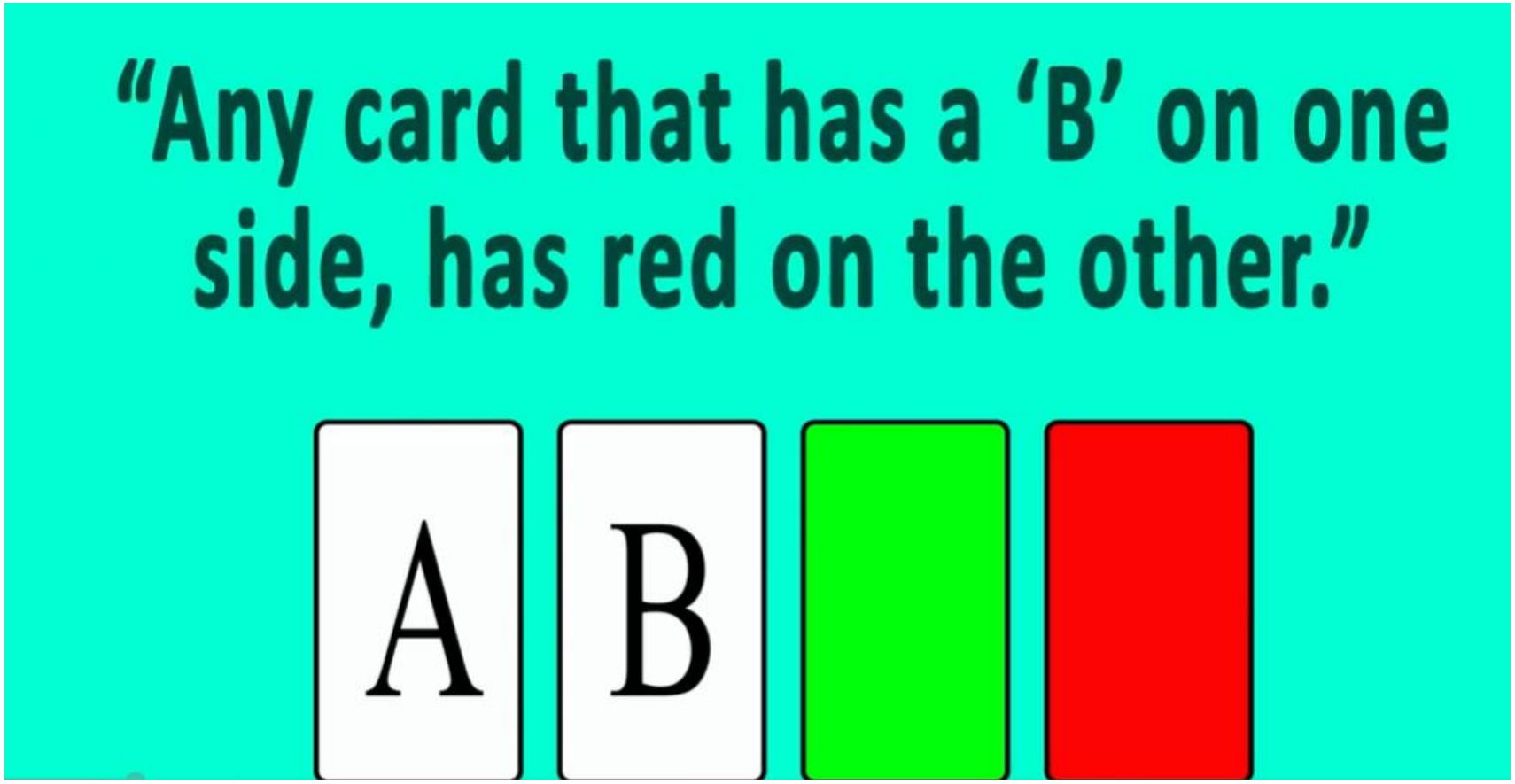
Inductive reasoning means: Drawing a general rule from repeated observations.

Example (Elephants)

- Every elephant you've seen → has a trunk
- Therefore → *all* elephants have trunks

Wason's Card Selection Task

Let’s have some play…



Abductive Reasoning

- Abduction is reasoning from an observed effect back to a possible cause.
- It's how we form best guesses or hypotheses to explain events we see.
- Unlike deduction (certain conclusion) or induction (generalizing), abduction is about inference to the most likely explanation—but it's not guaranteed to be true.

Example

- We know: Sam always drives fast when she's been drinking.
- Observation: Sam is driving fast.
- Abductive inference: Sam has probably been drinking.

Why is it Unreliable?

- Other reasons could explain Sam's speed (e.g., an emergency).
- The inferred cause is just a **plausible guess**, not a certainty.

Class Activity 1

https://tinyurl.com/Spring2026-Activity1

Problem solving

- **Gestalt theory of Problem Solving**
- **Problem space theory of Problem Solving**
- **Analogy in problem Solving**

Skill Acquisition

Skill acquisition is about how we **get better at familiar problems** over time. Unlike problem solving for a brand-new problem, skill involves **practice, memory, and pattern recognition**.

Anderson’s ACT Model

- **Declarative stage (learning rules)**
  - Use general-purpose rules, explicit memory, slow processing
  - Example: Learning to cook from a recipe book, step by step
- **Procedural stage (practice makes it easier)**
  - Convert rules into **specific procedures** for each case
  - Example: Knowing exactly how long to bake a casserole or cake
- **Generalization / Expert stage**
  - Extract **common patterns** → create general-purpose rules
  - Behavior becomes **automatic and fast**
  - Example: Experienced cook can adjust baking time intuitively for any casserole

Error & Mental Modes

Why do we make mistakes, and can we avoid them?

- Some errors result from changes in the context of skilled behavior. If a pattern of behavior has become automatic and we change some aspect of it, the more familiar pattern may break through and cause an error.

Example: We intend to stop at the shop on the way home from work but in fact drive past. Here, the activity of driving home is the more familiar and overrides the less familiar intention.

Mental Model Mismatch – Classic Example (Lift Case)

Emotion

- **James–Lange Theory**
  - Body reacts first → emotion is interpreted later
  - “We are afraid because we tremble”
- **Cannon’s View**
  - Body responses are too slow & similar → emotion ≠ only physiology
- **Schachter–Singer Theory (Most useful for HCI)**
  - Emotion = **physical response + situation interpretation**
  - Same heartbeat → excitement or fear based on context
- **Key takeaway for designers**
  - Emotion = biological + cognitive (called **affect**)
  - Affects how users handle interfaces

Example

- Before hackathon results → excitement
- App crashes during online exam → panic

Individual Differences

- Users are **not all the same**, even if general psychological principles apply
- **Types of individual differences**:
  - **Long-term**: gender, physical abilities, intellectual abilities, disabilities
  - **Short-term**: stress, fatigue, illness, distraction
  - **Change over time**: age
- Design decisions can **negatively affect or exclude** some users
- Good design should:
  - Support multiple sensory channels (visual, audio, haptic)
  - Avoid pushing users to cognitive or perceptual limits
  - Be usable even under pressure or distraction

Example Campus app / learning platform:

- Dark mode, captions, screen reader support → inclusive
- Simple navigation helps 1. Sleep-deprived students 2. Stressed users during exams 3. Users multitasking on mobile